## Bench

### 1 GC B cells



Peyer's patches; sort B220$^{+}$GL7$^{+}$Fas$^{+}$, ~10,000 cells per mouse

**steps 1–4 · 4–5 h**

### 2 Genomic DNA



proteinase K lysis; Wt and AID-null mice side by side

**steps 5–6 · 2 h**

### 3 $J_H4$ PCR



1.2-kb amplicon, high-fidelity polymerase; gel check, clean-up

**steps 7–10 · 3 h**

### 4 Sequencing



full-length amplicons, R10.4.1 flow cells; FASTQ per library

**steps 11–12 · 1–2 days**

## Analysis with SHMtrace

### 5 Project



sample sheet: file, sample, group, control; app or command line

**steps 13–15 · 15 min**

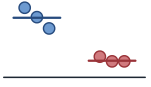
### 6 Run



read preparation, control-calibrated calling, animal-level tests

**steps 16–17 · 2–10 min**

### 7 Report



controls, margin, read accounting, warnings; spectra, maps, clones

**steps 18–23 · 30 min**

### 8 Plan



power from this analysis as a pilot: animals per group

**step 24 · 5 min**